

Haotian Xu 许皓天

RL algorithms and RL infrastructure for long-horizon agents — sole owner of the RL framework, RL training, and credit-assignment algorithms behind Apodex 1.0 and Apodex 1.1.

haotian@apodex.com htxu91@gmail.com yyht.github.io/personal-page 2,000+ citations · h-index 16 · i10 22

SUMMARY

RL researcher and infrastructure engineer working on reinforcement learning for agents that run for thousands of steps. I built the asynchronous training stack for both Apodex flagship releases end to end, designed the fragment-level and token-level credit-assignment algorithms (PIVOT-RL, SAO), and ran the RL that produced the 1.0 → 1.1 benchmark jump — on a 397B-class MoE flagship and a 35B-A3B open-weight Mini. Co-author of OpenRLHF and 10+ papers at NeurIPS, ICLR, ACL, EMNLP, and TPAMI.

RESULTS APODEX 1.0 → 1.1

The RL stage I owned is what separates the two releases. Against Apodex 1.0, the 1.1 flagship gains **+35.0 on FrontierScience-Research** (28.3 → 63.3), **+22.0 on APEX-Agents** (16.5 → 38.5, 2.3×), **+19.5 on GDPVal** (59.3 → 78.8) and **+14.0 on FrontierFinance** (40.3 → 54.3), with simulated-business net worth on YC-Bench rising **21.6×** (\$47,966 → \$1,038,255); HLE (49.0 → 56.1) and DeepSearchQA (84.6 → 92.4) moved up too, confirming the agentic gains cost nothing in general reasoning or search. The 35B-A3B open-weight Mini reproduces the pattern on the same recipe at a twelfth of the scale (FrontierScience-Research +26.7, FrontierFinance +17.0, APEX-Agents +12.3), which is the evidence that the recipe rather than the parameter count is carrying the result. Artificial Analysis independently ranks Apodex 1.1 at **Intelligence Index 44, #11 of 177 models** against a median of 18, with GDPval-AA v2 Elo 1348 — ahead of DeepSeek V4 Pro and Qwen3.7 Max.

CONTRIBUTIONS INFRASTRUCTURE

Fully-asynchronous agentic RL framework

Designed and built the training framework for both releases on top of miles, against the defining pathology of agentic RL: trajectory wall-clock cost is wildly irregular across search, code execution, file processing, and multi-agent coordination, so any synchronous pipeline idles on its slowest episode.

- **Fully decoupled loops** — training workers and rollout engines run on independent loops, communicating only through a shared data buffer and an output queue. Completed trajectories enter optimization asynchronously, with no barrier on stragglers.
- **Non-blocking evaluation pool** — a dedicated pool scores checkpoints without ever interrupting the training loop, taking evaluation off the critical path.
- **Hybrid-colocate GPU reclamation** — idle training cards are recruited as generation engines, converting bubble time into rollout throughput.

CREDIT ASSIGNMENT FRAGMENT

PIVOT-RL: tree-policy fragment selection

The headline RL algorithm of Apodex 1.1. A long agentic rollout terminates in a single 0/1 verifier reward: terminal outcomes give only coarse supervision, useful intermediate work routinely precedes a late failure, and successful traces still contain weakly-grounded decisions. PIVOT-RL makes the **trajectory fragment** the unit of credit — and picks which fragment to train on by tree search rather than by a heuristic cut.

- **Tree-policy pivot selection** — expand an on-policy search tree from the model's own rollout instead of cutting one linear trajectory after the fact. Sampling continuations from each candidate fork gives Q-value estimates at the branch, so the pivot is chosen on measured branch value. This costs more rollout compute

than a heuristic cut and buys a markedly more reliable pivot-turn choice on long-horizon tasks.

- **Branch-point detection** — candidate forks come from first-error-step detection, high-entropy decision points, and offline LLM-judge screening for bad patterns, targeting the places where the policy starts following an unproductive strategy, leans on insufficient evidence, misuses a tool, or fails to revise an assumption.
- **Scaling** — on three held-out agentic evaluations (search, knowledge, science), scores improve consistently from 20 to 140 rollout steps.

Write-up — [Credit assignment after the terminal reward](#)

CREDIT ASSIGNMENT CRITIC

Critic pretraining: HL-Gauss loss and reasoning-trace warmup

A randomly-initialized value head makes every early advantage noise, and published single-rollout async PPO trains its critic with plain MSE. In *miles-values* I replaced that regression objective with classification and warm-started the critic on reasoning traces.

- **HL-Gauss value loss** — the value head emits `num_bins` logits over a categorical support; the scalar target is projected onto a Gaussian-smoothed distribution ($\sigma = 0.75 \times \text{bin width}$, ~6 bins of supervision) and trained with cross-entropy. Shipped at 51 bins over $[0, 1]$. Bounded softmax gradients tolerate noisy targets, and GAE is unaffected — it still reads a scalar $V = \text{softmax}(\text{logits}) \cdot \text{support}$.
- **Reasoning-trace warmup** — pretrain the critic on trajectories from earlier RL runs, so it enters PPO calibrated instead of climbing out of deeply negative explained variance.
- **IID pretrain/RL data beats more data** — with the critic corpus overlapping the RL set, EV holds ≈ 0.2 throughout and both AIME24/25 gain further; on a disjoint corpus the advantage decays back to the un-pretrained curve. Cheapest recipe: roll the critic corpus from the same SFT-filtered data used for RL.
- **Math results** — Qwen3.5-4B / deepmath-103k: HL-Gauss > MSE (MSE's AIME24 flat after 200 steps). Qwen3.5-35B-A3B / dapo-17k: all variants reach AIME24 0.91 / AIME25 0.85, but sample efficiency orders `hl-gauss-pretrain` > `hl-gauss` > `MSE`.
- **Agentic results** — 128k search-agentic task, single-rollout PPO on a deliberately small critic warmup (60–70k traces, 10 steps). By step 129 against baseline:
BrowseComp 58.0 → 70.0 (+12.0), FrontierScience 45.0 → 53.3 (+8.3), HLE 38.0 → 46.4 (+8.4), with APEX-Agents held flat (43.2 → 42.9).

Write-ups — [SAO, part 1](#) · [SAO, part 2](#) · [value pretraining](#)

RELIABILITY

Diagnostics for multi-week RL runs

Owned the instrumentation that kept 397B RL runs honest: loss-mask-aware explained variance aggregated across microbatch/DP/CP (the metric separating a learning critic from a silently dead one), rollout-structure collapse detection (parallel multi-query degradation under penalty-only rewards with length-normalized objectives), reward-hacking audits, benchmark-leakage closure for SWE-bench Verified-class evaluations, and FP8-vs-BF16 inference parity checks.

EXPERIENCE

Apodex *RL Algorithms & RL-Infra, Agentic RL*

2025.11 — Present

Formerly MiroMind. Sole owner of the RL framework, RL training, and credit-assignment algorithms for Apodex 1.0 and Apodex 1.1. Built the fully-async training stack, PIVOT-RL and SAO credit assignment; drove the 1.0 → 1.1 gains above on both the 397B flagship and the 35B-A3B open-weight Mini.

Xiaohongshu — HiLab *Math Reasoning & Agentic RL*

2024 — 2025.8

RL for mathematical reasoning and tool-using agents; agentic RL scaling laws for spontaneous code execution (NeurIPS 2025).

OPEN SOURCE	MILES-VALUES	Dedicated SAO (Async Single-rollout PPO) framework with large-scale value pretraining.
	ASYNC_OPENRLHF	Fully-asynchronous RLHF pipeline extending OpenRLHF.
	AGENTHARNESS	Public evaluation harness reproducing Apodex-1.0 ReAct results across nine public deep-research benchmarks.

SELECTED PUBLICATIONS	Apodex 1.1: Scaling Agentic Intelligence for Complex Work	arXiv:2608.23283 · 2026
	Apodex-1.0: A Verification-Centric Agent Team for Discoverative Intelligence	Apodex Tech Report · 2026
	From System 1 to System 2: A Survey of Reasoning Large Language Models 529 cites	IEEE TPAMI 2025
	OpenRLHF: A Ray-based Easy-to-use, Scalable and High-performance RLHF Framework 371 cites	EMNLP 2025
	Agentic RL Scaling Law: Spontaneous Code Execution for Mathematical Problem Solving 71 cites	NeurIPS 2025
	IterResearch: Rethinking Long-Horizon Agents with Interaction Scaling 44 cites	ICLR 2026
	Search Self-Play: Pushing the Frontier of Agent Capability 25 cites	ICLR 2026
	GEM: A Gym for Generalist LLMs 17 cites	ICLR 2026
	Omni-DPO: A Dual-Perspective Paradigm for Dynamic Preference Learning 9 cites	ICLR 2026
	TL;DR: Too Long, Do Re-weighting for Efficient LLM Reasoning Compression 10 cites	ACL 2026
	Shuttle Between Instructions and Parameters	ACL 2026

EDUCATION	Tsinghua University	Department of Electronic Engineering	2013 – 2016
	UESTC	B.Eng.	2009 – 2013

TECHNICAL PROFILE	RL ALGORITHMS	PPO/GRPO and group-normalized variants, GAE, value pretraining, off-policy importance correction and token-level clipping, fragment- and turn-level credit assignment, RLHF, MCTS, DPO variants, reward modeling and anti-reward-hacking.
	RL INFRASTRUCTURE	Fully-async rollout/training decoupling, hybrid-colocate GPU reclamation, shared-buffer pipelines, Megatron and torch-dist checkpointing and conversion, TP/CP/EP parallelism, vLLM and SGLang rollout engines, FP8 inference parity, Ray, OpenRLHF, miles.
	LANGUAGES	Chinese (native), English (professional).